

Public Attitudes Toward Marginalized Groups’ Online Self-Disclosure: A Statistical Study of ”Coming Out” Narratives on Bilibili

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Abstract

This report aims to investigate the characteristics and challenges faced by LGBTQ+ individuals when expressing themselves online, as well as to analyze public commentary on such expressions. Using Bilibili as the data source and ”coming out” as the keyword, we collected relevant video information and selected representative videos to extract associated comments. Data analysis methods including word clouds, keyword analysis, paired analysis, and co-occurrence network analysis were employed to examine the perspectives of narrators and observers. The study identifies four key findings: (1) The absence of the ”father” role as a listener in the coming-out process; (2) Gender differences in self-expression: lesbian individuals tend to remain silent compared to gay men; (3) Observers’ evaluation differs by gender: male and female coming-out narratives attract attention to different aspects, namely ”offspring” and ”marriage,” respectively; (4) Younger audiences tend to accept and support LGBTQ+ communities, whereas older audiences hold more conservative views. Based on these findings, the report proposes strategies to help LGBTQ+ individuals overcome challenges and to foster mutual understanding and communication between LGBTQ+ and heterosexual communities.

Keywords: LGBTQ+, Bilibili, observers, narrators, comments, video data

1 Introduction

1.1 Overview: LGBTQ+ Communities as ”Marginal Persons”

The concept of the ”marginal man” was introduced by the American sociologist Robert Ezra Park (1928) in his studies of urban immigrants in the United States. Building on the notion of the ”stranger” defined by German philosopher Georg Simmel (1950), Park argued that a stranger appears ”strange” because participation in a new social environment leads to a lack of cultural identification. This results in a perceivable lack of closeness between two or more individuals or social categories. A ”marginal man” refers to someone whose participation in two social groups is incomplete, thus occupying the cultural periphery of both.

Throughout long processes of social evolution, LGBTQ+ communities have gradually developed self-identity and social structures while simultaneously being influenced and challenged by mainstream culture. Over time, these communities have formed their own collective identity, behavioral norms, value systems, symbolic codes, and discursive practices. Compared with the dominant heterosexual culture, LGBTQ+ individuals, in Park’s view, are groups that seek recognition yet awkwardly navigate outside mainstream acceptance, thereby exhibiting the characteristics of a subculture. The role of the ”marginal person” in the interaction and integration between this subculture and mainstream culture deserves further study [1].

Globally, the LGBTQ+ population has increased from approximately 50 million in 1830 to around 400 million in 2023 (Figure 1). However, they still account for only about 3% to 5% of the total population, remaining a socially marginalized group. One study abroad [2] indicates that the proportion of gay and lesbian individuals is roughly similar, with women slightly higher at 4.2% compared to men by about 1% (excluding bisexuals and other uncertain factors, Figure 2).

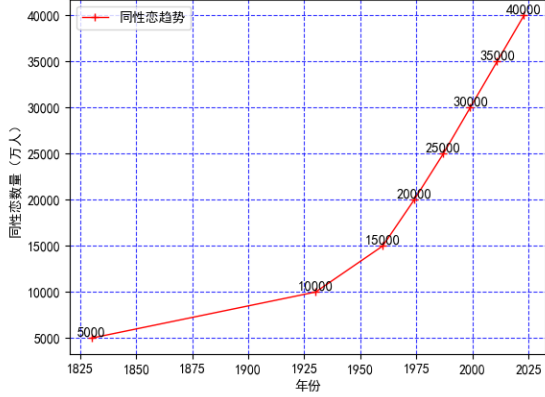


Figure 1

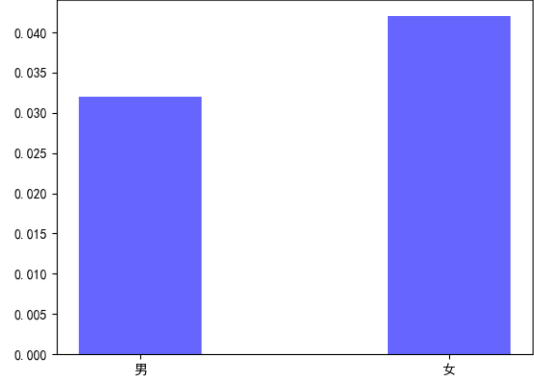


Figure 2

In China, according to the 2020 International LGBTQ+ Commission survey, the total LGBTQ+ population has surpassed 70 million. Cities with the largest LGBTQ+ populations include Chongqing, Shanghai, Beijing, Chengdu, Guangzhou, Shenzhen, Wuhan, Tianjin, Xi' an, Hangzhou, Hong Kong, Zhengzhou, Nanjing, Shijiazhuang, Shenyang, Dongguan, Changsha, Suzhou, Harbin, and Hefei. Despite these numbers, due to China's large population base, LGBTQ+ individuals still account for only about 5% of the population. This report also provides an interactive provincial map of LGBTQ+ population distribution (see Appendix).

1.2 Why Focus on LGBTQ+ Communities?

As a socially marginalized group, LGBTQ+ individuals often receive insufficient social support. Levels of public acceptance vary widely across countries: according to global statistics, Syria has the lowest acceptance rate at under 1%, while Norway has the highest at 92% (an interactive world map of LGBTQ+ acceptance is included in the Appendix). Notably, no country has achieved a 100% acceptance rate, indicating that the social environment for LGBTQ+ individuals remains far from fully welcoming.

Research shows that 56.9% of adolescent lesbians in China report symptoms of depression, significantly higher than the 36.94% reported among the general female population, and also higher than the 37.8% depression rate reported among MSM in a study conducted in Mianyang.

With the rapid development of the internet and the rise of various livestreaming and video-sharing platforms, it has become easier for the public to access information about groups outside their own social circles. In this digital era, LGBTQ+ individuals increasingly gain opportunities to voice their experiences, while the internet provides them with a platform to present their authentic selves, share positive emotions, foster integration into mainstream society, and build independence [3].

However, increased visibility also raises the risk of being submerged within the overwhelming flow of mainstream information. For this reason, it is crucial to pay sustained attention to LGBTQ+ groups and to their patterns of expression and behavior.

2 Data Description

Guided by the research framework, this study aims to identify the characteristics and challenges faced by sexual minority individuals when engaging in online self-disclosure, as well as to analyze public reactions to such disclosures. Based on this objective, three major social media platforms—Weibo, Douban, and Bilibili—were initially considered for data collection.

Preliminary exploration revealed that most Weibo posts containing the keyword “coming out” were related to celebrities or fan communities, resulting in low data validity and substantial noise. Douban, meanwhile, contained a high proportion of dating-related content and relatively few posts sharing personal disclosure experiences. In contrast, Bilibili featured a substantial number of videos in which users narrate their own “coming out” experiences, and the associated comments displayed relatively high textual quality. Considering data relevance and reliability, Bilibili was therefore selected as the data collection platform for this study. The final dataset consists of two components: video-level information and comment-level information.

2.1 Video Data

Using the Octopus web-scraping tool, we retrieved 839 videos tagged with the keyword “coming out,” collecting metadata including video titles, view counts, number of comments, duration, and upload time. Variable descriptions are provided in Table 1. After manually removing videos unrelated to actual disclosure narratives, 715 videos remained, yielding an effective data rate of approximately 85%. Engagement metrics such as likes, coins, favorites, and view counts were recorded as integer variables.

2.2 Comment Data

Scraping comments from all 715 valid videos would result in an excessively large dataset. Therefore, we selected six representative videos from the top-ranked search results. To ensure comprehensiveness, the selected videos included both successful and unsuccessful coming-out experiences, as well as content narrated from different perspectives (e.g., parent-centered versus child-centered narratives). The titles and view counts of the selected videos are shown in Table 3.

All comments from these six videos were collected, resulting in a total of 5,626 valid comments. Variable descriptions for the comment dataset are provided in Table 2.

3 Narrator-Side Analysis

This section focuses on the video-title text data retrieved from Bilibili. We begin with numerical feature extraction to obtain an overall understanding of the characteristics of “coming out” videos. We then conduct keyword extraction and paired-term analysis to identify the roles of disclosure recipients and narrators. From this micro-level family perspective, we highlight several difficulties sexual minority individuals face during the coming-out process.

3.1 Numerical Features of “Coming Out” Videos

First, we examine the distribution of view counts across the 715 cleaned and validated videos. Records containing missing values or invalid strings were corrected prior to analysis. Because the magnitude of view counts varies substantially, we apply a logarithmic scale on the vertical axis when constructing the histogram. The distribution is heavily right-skewed, and overall view counts remain low, suggesting that most self-disclosure videos posted by sexual minority individuals receive limited public attention.

Next, we compute viewer engagement rates by dividing likes, coins, favorites, and bullet comments by view counts for each video, then visualize these indicators using boxplots. In terms of feedback intensity, likes exceed favorites, which in turn exceed coins, with bullet comments being the least common form of engagement. Like-rates appear substantially higher than other types of feedback, indicating that “liking” is the primary mode of user interaction.

We then examine upload-time patterns across multiple temporal resolutions—monthly, weekly, and hourly—to identify potential posting behaviors of narrators. Monthly data show mild fluctuations in upload volume. At the weekly level, narrators appear less likely to post in the middle of the week, which is consistent with corresponding monthly trends. At the intra-day level, posting frequency peaks around mealtimes, while video uploads gradually increase throughout the daytime. This pattern suggests that narrators may intentionally align their posting behavior with user viewing habits.

Finally, we analyze video length. Most videos fall within the short-to-mid duration range, and longer videos tend to receive fewer views. This pattern suggests that concise, well-edited videos may help increase visibility and attract greater attention for this marginalized group.

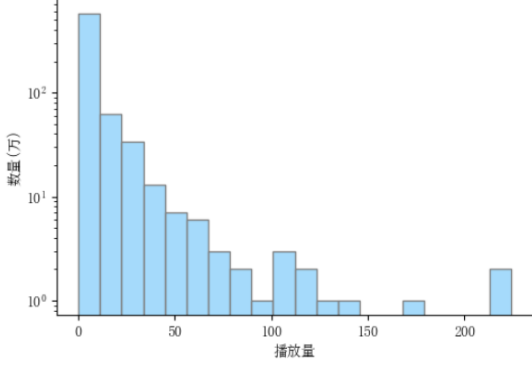


Figure3: View Counts for "Coming Out" Videos

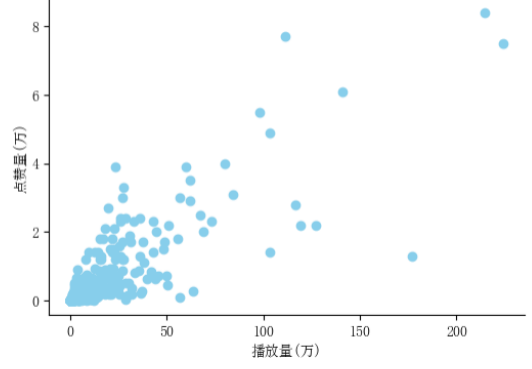


Figure 4:View Count vs. Like Count

3.2 Word Cloud and Keyword Analysis: Confidants and Speakers

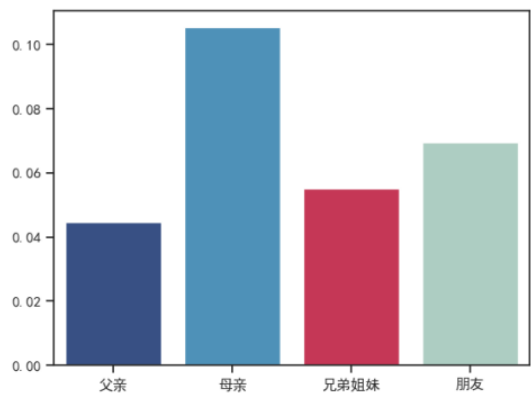
Beyond numerical features, the textual content of video titles also contains rich information, such as the characteristics of the speaker and the intended confidant. After identifying the quantitative patterns of "coming out" videos, this section further analyzes the title texts to answer two key questions: *Who is speaking?* and *To whom are they speaking?*

First, we conducted word segmentation on all video titles. The titles were converted into a single string and segmented using the `jieba` package, followed by a stop-word removal step to filter out non-informative terms. A word cloud was then generated based on the processed text. As shown in the figure, the word cloud highlights several major themes in speakers' narratives: family-related terms such as "parents," "mother," and "younger brother"; psychological terms such as "forced" and "confession"; and broader social terms such as "marriage." Overall, the word cloud reflects the anxiety, internal conflict, and emotional tension that sexual minority individuals experience when "coming out" to their families, particularly their parents.

Next, we extracted keywords using the `jieba.analyse.extract_tags()` function based on the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm, a widely used method in text mining for evaluating how important a word is within a document. After extracting the top 150 words with the highest TF-IDF scores, we examined the weights of several sets of words representing different relational categories: "father" (including variants such as "dad"), "mother" (including "mom"), "siblings" (e.g., "older brother," "older sister"), and "friends" (including "classmates," "roommates"). These aggregated weights were visualized using a bar chart (Figure 12).

Regarding the intended confidant, Figure 12 shows that references to "father" are significantly less frequent than references to "mother." The aggregated TF-IDF score for words related to "father" is only around 0.04, compared to 0.11 for "mother." This suggests that speakers tend to communicate or "confess" to their mothers rather than their fathers during the coming-out process. The relative absence of the father figure may therefore constitute an important barrier in the coming-out experience. This pattern is further reinforced in the comment analysis presented later.

In terms of the speakers themselves, the results indicate that slightly more gay men than lesbian women appear among the "coming out" videos. This aligns with the broader social reality that "gay men" receive more public attention and discourse than "lesbian women." Compared with gay men, lesbian individuals appear more "silent." One possible explanation is that lesbian women, as women, may carry heavier social expectations related to marriage and childbirth, making them more reluctant to speak out. Additionally, within the broader LGBT movement and public discourse, lesbian-specific issues have received insufficient attention and recognition, rendering lesbian communities comparatively more "invisible" than gay male communities.



4 Bystander Perspective: Analysis of Comment Data

This section focuses on the scraped comment texts. We begin with sentiment analysis and a word cloud to obtain an initial understanding of public attitudes, represented by Bilibili users. We then conduct keyword analysis and text co-occurrence network analysis to identify the major themes emphasized by commenters in their discussions of sexual minority individuals' coming-out experiences.

4.1 Sentiment Analysis: Public Attitudes

We first applied Alibaba Cloud’s open sentiment analysis API to classify the emotional polarity of each comment, categorizing them into “positive,” “neutral,” and “negative” labels. As shown in Figure 15, positive comments constitute the majority, accounting for 74.54% of all comments. In contrast, 23.41% were predicted as negative, and 2.05% as neutral. Overall, these results suggest that bystanders on Bilibili tend to hold generally positive and accepting attitudes toward sexual minority groups.

Next, we processed all comment texts to generate a word cloud. All comments were concatenated into a single string and segmented using the `jieba` package. Stop-word filtering was applied to remove non-informative terms. Consistent with the sentiment analysis results, many frequently occurring words exhibit positive emotional valence, such as “support,” “hope,” “happiness,” “accept,” and “stay strong.” Among them, “accept” appears 660 times and “support” appears 464 times.

In addition, the word cloud reveals key topics commonly emphasized by commenters. These include family-related issues (e.g., “parents”) and broader social issues (e.g., “marriage fraud”). Moreover, as shown in the word frequency analysis (Figure 7), first-person pronouns such as “myself” and “we” occur at relatively high frequencies, suggesting that some commenters are themselves members of the sexual minority community. Thus, in addition to capturing bystander attitudes, the comment texts also partially reflect shared experiences within the LGBTQ+ community.

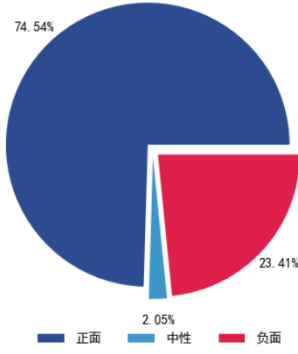


Figure6: Sentimental Analysis

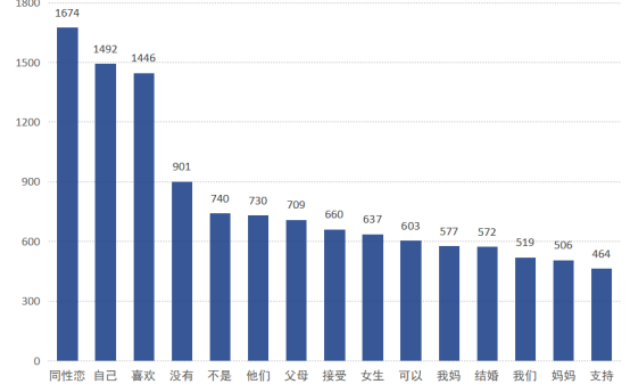


Figure 7: Word Frequency

4.2 Keyword Analysis: Shared Identity Characteristics

Building on the word cloud analysis, which preliminarily highlighted the importance of “father” and “mother” roles in the coming-out process, this subsection conducts a more systematic keyword-based examination. We first constructed a dictionary of identity-related keywords, grouping terms referring to “father” and “mother” respectively. We then iterated through all comment texts and calculated the frequency with which each identity category appeared. The results are shown in Figure 8.

As illustrated in the figure, references to the “father” role appear far less frequently than references to the “mother” role, accounting for only 19.28% compared to 29.29%. This suggests that in the context of coming out, both sexual minority individuals and commenters tend to emphasize communication with mothers, while intentionally or unintentionally neglecting fathers. Based on this pattern, we argue that the symbolic and practical “absence of the father” may constitute a significant barrier during the coming-out process for LGBTQ+ individuals.

4.3 Text Co-Occurrence Network Analysis: What Do We Talk About When We Talk About Homosexuality?

To complement the keyword frequency results—which primarily highlighted identity-related terms—this section applies a text co-occurrence network analysis to more comprehensively capture bystanders’ attitudes in the comment corpus.

Text co-occurrence network analysis is a semantic network method based on the assumption that if two words repeatedly appear within the same textual unit, they are semantically associated; in graph-theoretic terms, their corresponding nodes form an edge. In this study, each comment was tokenized, and word frequencies were recorded as node weights. Within each comment, all co-occurring word pairs were counted and stored as edge weights. The resulting node and edge files were exported as CSV and visualized using Gephi (see Figure 19).

To ensure readability, nodes with weights below 80 and nodes with degree zero were removed, leaving 27 nodes and 58 edges. Modularity detection identified three communities. Each community contains a salient identity-related term—“妈妈” (mother), “男生” (male), and “女生” (female). In the “female” community, the node “结婚” (marriage) exhibits a high weight, while in the “male” community, the node “孩子” (children) is strongly connected to other nodes. This suggests gendered differences in how bystanders evaluate “coming out”: discussions about gay men tend to center on lineage and having children—likely influenced by cultural expectations of patrilineal continuity—whereas discussions about lesbian women emphasize marriage and future life trajectories.

Across both the “male” and “female” communities, terms such as “支持” (support) and “希望” (hope) appear frequently, indicating generally positive attitudes toward individuals who come out. In contrast, the community centered on “妈妈” (mother) shows strong connections to negatively valenced words such as “恶心” (disgusting). This highlights the continued centrality of family tension in coming-out narratives, with the mother figure emerging as a key point of emotional friction. Notably, the lesbian-related colloquial term “拉拉” appears in this community, whereas no parallel term for gay men is present—suggesting that the “daughter-mother” communication chain may be particularly salient, and that mothers may be more likely to receive disclosures from daughters than from sons.

Taken together, the co-occurrence network reflects the complexities and challenges sexual minorities face when coming out within family contexts.

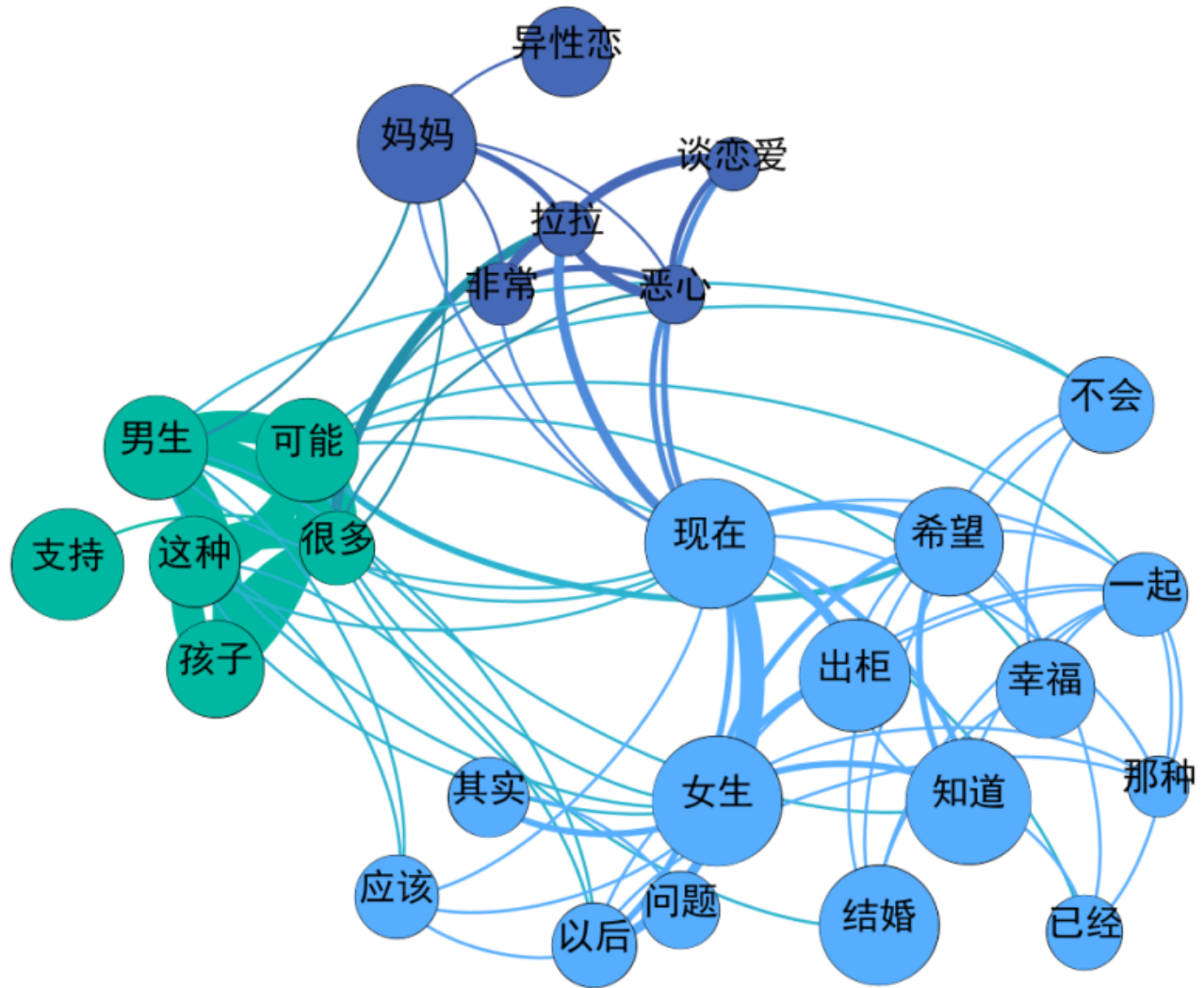


Figure 8: Text Co-Occurrence Network Analysis

5 Conclusion

Based on the analyses above, this report identifies the following key issues:

1. The **marginality** of the homosexual community;
2. **Recipient of disclosure**: the role of the father is notably absent during the coming-out process;
3. **Disclosure subject**: lesbian women tend to be more silent compared to gay men;
4. **Observers**: the focus of observers differs when evaluating male versus female homosexuals' coming-out experiences, centering on *offspring* for men and *marriage* for women;

5. Younger generations tend to be more accepting, whereas older generations hold more conservative views.

The rapid economic development has promoted social progress. Efforts in sex minority education and rights protection have gradually changed public perceptions and attitudes toward sexual minorities, reducing discrimination. According to surveys, by 2020, homosexuals accounted for only around 5% of China's total population. Although societal acceptance has improved in recent years, sexual minorities still tend to disclose their identity sparingly and follow certain patterns.

This study focuses on the “coming-out” community on Bilibili. Using numerical feature extraction, keyword analysis, and web-scraping scripts, we collected videos, comments, and on-screen bullet comments related to “coming out.” Through textual analysis, data integration, and keyword comparison, we analyzed the sentiment and content expressed by the community.

The main users of Bilibili are aged 16–35. They grew up in a rapidly developing social and cultural environment, which is more inclusive, and are generally less discriminatory toward sexual minorities. Despite this, coming-out videos by homosexual individuals generally have lower exposure, fewer likes, and fewer views compared to other types of content. Most videos are uploaded in the evening, indicating that homosexual individuals remain in a relatively marginal position in society. However, due to celebrity effects or dramatic cases, some videos still gain considerable attention. Analysis of 715 cleaned comments and bullet comments shows that most feedback comes from fellow sexual minorities offering congratulations, or from non-sexual minority users providing acknowledgment and understanding. Therefore, Bilibili users appear generally supportive and accepting of sexual minorities.

Nevertheless, older generations, particularly those born in the 1960s and 1970s, still hold more conservative views due to traditional cultural influences. These individuals often lack understanding of sexual minorities, and their attitudes can result in discrimination and social pressure. Family-related stress is particularly prominent; many homosexual individuals may marry someone of the opposite sex due to family pressure, giving rise to issues such as “tongqi/tongfu” (married partners of gay men and lesbian women). Data indicates that sexual minorities tend to disclose their identity first to their mothers, as mothers are culturally associated with a more compassionate and accepting role within the family. Fathers, conversely, are often less receptive, which can create intergenerational tension and, in some cases, family breakdowns. Historically, homosexual individuals were stigmatized, and homosexuality was even classified as a mental disorder in China before 2001, contributing to long-term social discrimination.

In terms of gender, male homosexuals disclosed slightly more frequently than female homosexuals, despite surveys showing that lesbian women account for a slightly higher proportion of the population. This disparity reflects traditional cultural constraints on women, leading female sexual minorities to conceal their identity more often. Moreover, metrics such as video exposure, likes, and views show that sexual minorities remain marginalized in terms of social discourse, even on a relatively tolerant platform like Bilibili.

Furthermore, the preference to disclose to mothers reflects the mother's cultural role as a compassionate and accepting figure in the family. Finally, when discussing sexual minorities, Bilibili users generally exhibit supportive attitudes. However, older individuals still prioritize family continuity and reputation, placing limits on personal expression. Therefore, male sexual minorities are encouraged to be more vocal in order to strengthen their voice and gradually improve public awareness and recognition, eventually extending greater social support to female sexual minorities.

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